

HACKCULT '26 · PS2

HeritageLens

Rediscovering India's architectural memory through AI

Digital revitalisation of unstructured cultural and architectural heritage archives

TEAM HERITAGELENS · IIT GANDHINAGAR

Heritage knowledge lives in disconnected formats

Photographs, PDFs, field journals and regional-language scans cannot be searched as one archive.



PHOTOGRAPHS

Visual evidence without structured context



FIELD NOTES

Observations trapped in isolated records



SCANNED DOCUMENTS

Text remains invisible to search

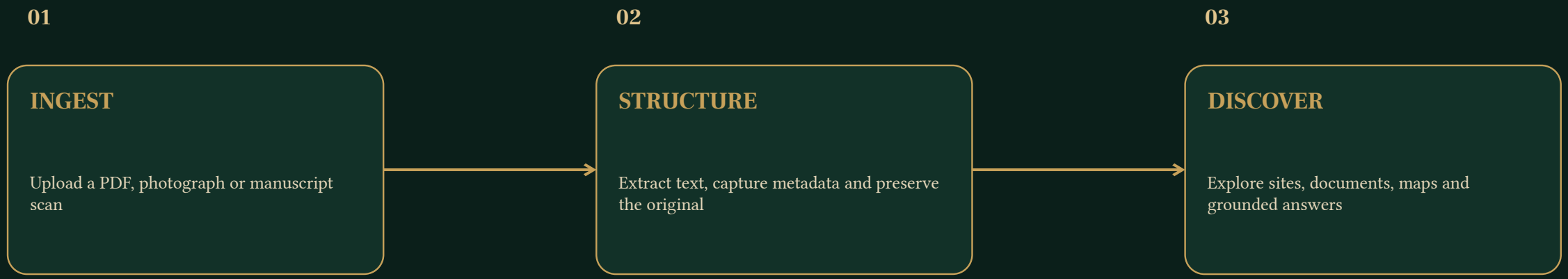
THE CONSEQUENCE

Discovery slows down.
Context breaks.
Fragile sources stay hidden.

Researchers cannot trace evidence across collections without manually reopening every source.

One archive connects source files to cultural context

HeritageLens turns unstructured evidence into records people can browse, search and verify.



Public archive

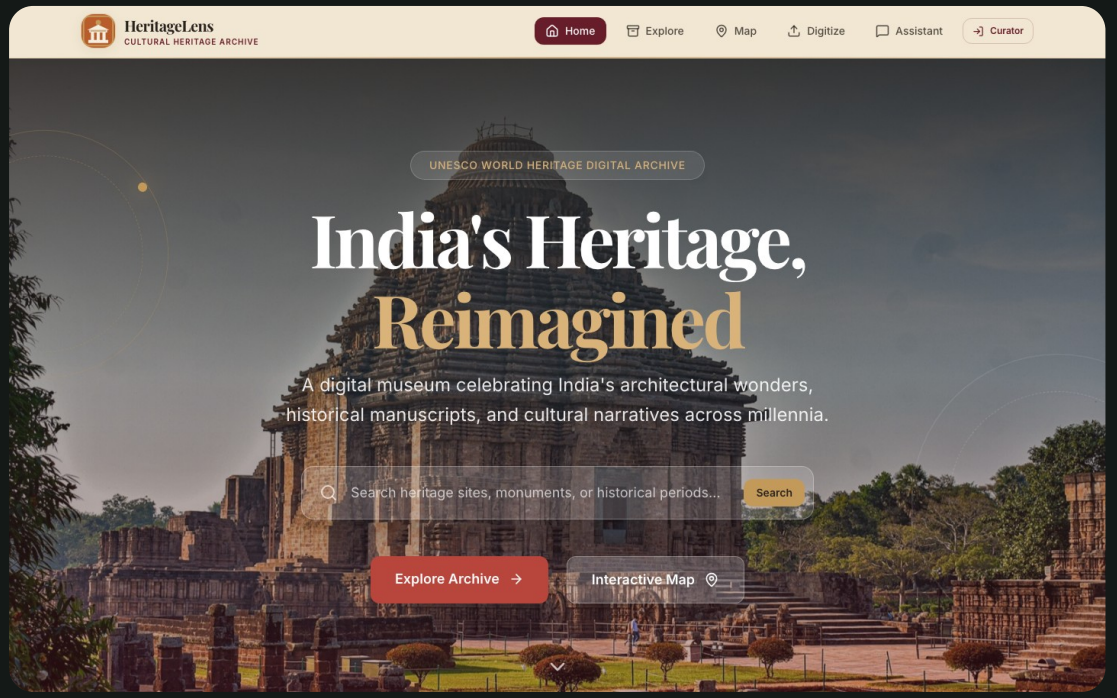
Browsing, filters, map and document search remain open to everyone.

Curator workspace

Authenticated upload, review, correction and archive maintenance.

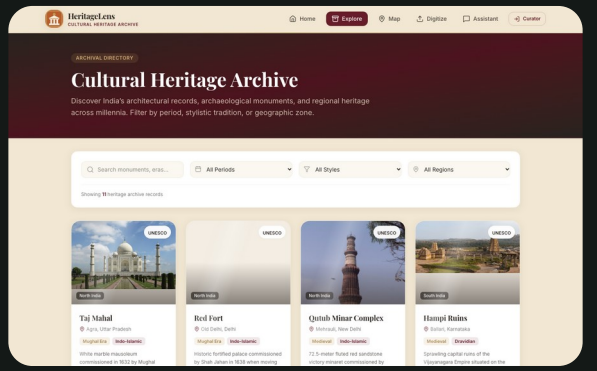
The working product

Live screens from heritage-pearl.vercel.app. No mock interface is presented as real.



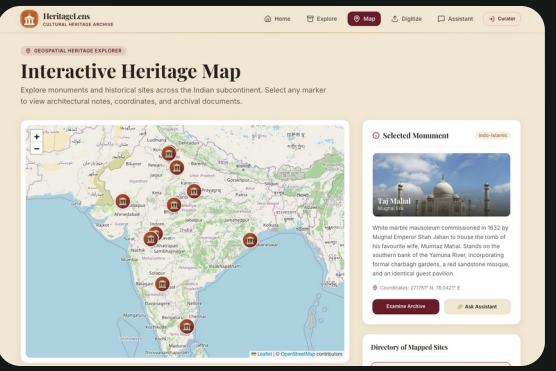
IMMERSIVE ENTRY

Search and visual discovery



SEARCHABLE ARCHIVE

11 seeded heritage records with filters and document search



GEOSPATIAL EXPLORER

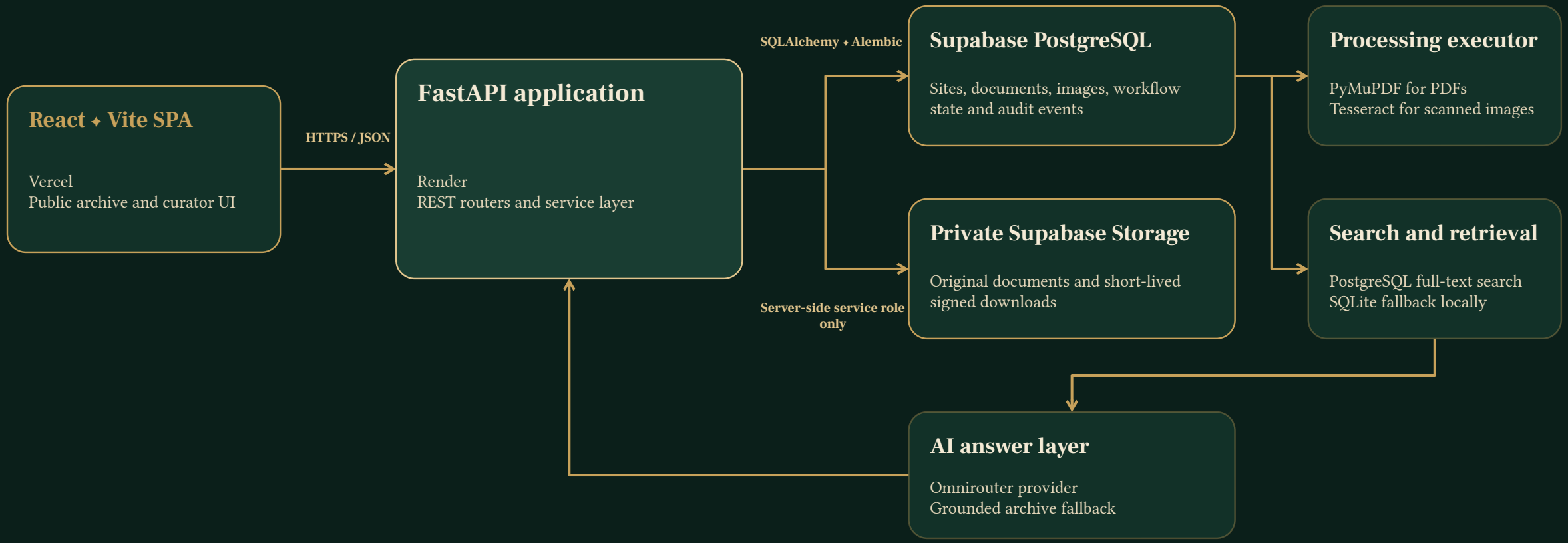
Interactive markers connect monuments to archival detail

Also implemented

Curator login · OCR upload · review workflow · grounded archive assistant

System architecture

A modular monolith keeps the hackathon build coherent while preserving clean service boundaries.



Data and archival architecture

The schema preserves cultural context, original evidence and the complete processing lifecycle.



Document digitization workflow

Automation handles extraction. A curator controls what enters the searchable archive.

AUTOMATED

HUMAN REVIEW

01

Upload

PDF or image

02

Validate

Type and size

03

Store

Private object key

04

Extract

PyMuPDF or OCR

Review and correct

Curator verifies text and metadata

Searchable archive

Approved text becomes discoverable

PERSISTED STATES

UPLOADED

PROCESSING

EXTRACTED

NEEDS REVIEW

APPROVED

FAILED

Search and source-grounded answers

Search reaches both site metadata and curator-corrected document text.

IMPLEMENTED

User question

Natural language or archive query

Indexed archive

Site fields plus extracted and corrected text

Evidence first

Matching records, citations and grounded synthesis

PostgreSQL full-text search in production
Portable substring fallback for SQLite development

Omnirouter when configured
Grounded archive fallback when unavailable

PHASE 2 ROADMAP

Multilingual semantic retrieval

Embeddings and hybrid ranking across regional-language archives.

RAG evaluation

Measure retrieval quality and citation faithfulness before expanding the corpus.

Engineering choices that support growth

The current architecture favors tested boundaries and reversible deployment decisions.

40% — TECHNICAL EXECUTION

Protected write APIs, explicit processing states, audit trail, migration-managed schema

30% — ARCHITECTURE

Modular monolith, durable object storage, PostgreSQL indexes, configurable providers

20% — UI / UX

Public discovery, curator review flow, direct-route support, meaningful failure states

10% — PROBLEM FIT

OCR, correction and source-grounded retrieval for unstructured heritage archives

VERIFIED: 39 backend tests · 3 browser acceptance workflows · frontend lint and production build

A living digital record of India's architectural memory

HeritageLens gives every source a durable place, searchable context and a path back to evidence.

Researchers

Find connected evidence across sites and documents

Archivists

Preserve originals while correcting machine-extracted text

Institutions

Publish structured collections through a maintainable system

Heritage tourism

Explore monuments through maps and accessible narratives

Technology should help cultural memory remain discoverable, verifiable and alive.

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